

Digital Steganography

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Outline

- What is Steganography?
- Uses of Steganography
- Its Relation to Cryptography
- Digital Watermarking
- Network Security and Steganography
- Types of Steganography
- Detecting Steganography: Steganalysis
- Future of Steganography

What is Steganography?

- Origin of the term: *steganos*: secret – *graphy*: writing



History of Steganography

- Steganography in Ancient Greece
- Ancient Rome - 1499, *Steganographia*, the first book about steganography
- World War II Germany - microdots
- Modern Steganography - 1985, with the development of personal computers
- Digital Steganography – with internet (*e.g. digital watermarking*)

Uses of Steganography

- **Governments**
- **Businesses:** *Digital Watermarking*
- **Individuals**

Steganography & Cryptography

- Steganography and Cryptography are closely related
- The difference is in their goals...
 - Cryptography: although encrypted and unreadable, the existence of data is not hidden
 - Steganography: no knowledge of the existence of the data
- Steganography and Cryptography can be used together to produce better protection

Digital Watermarking



Image “painted” with the watermark: “Invisible Man” © 1997, Neil F. Johnson

Digital Watermarking

- Used primarily for identification
- Embedding a unique piece of information within a medium (typically an image) without noticeably altering the medium
- Almost impossible to remove without seriously degrading an image

Digital Steganography & Watermarking

- Digital watermarking hides data in a file, and the act of hiding data makes it a form of steganography
- The key difference is their goals...
 - Steganography: hiding information
 - Watermarking: extending the file with extra information
- Steganographic information must never be apparent to a viewer unaware of its presence.

Network Security and Steganography

- **Confidentiality:** *any unauthorized person does not know that sensitive data exists*
- **Survivability:** *verifying that the data does not get destroyed in the transmission*
- **No detection:** *cannot be easily found out that the data is hidden in a given file*
- **Visibility:** *people cannot see any visible changes to the file in which the data is hidden*

Types of Digital Steganography

- **Hiding a Message inside Text**
- **Hiding a Message inside Images**
 - Most popular technique
- **Hiding a Message inside Audio and Video Files**

Hiding a Message inside Text

- **Partially effective**

random capitalosis is a rare disease often contracted by careless internet users. this sad illness causes the affected person to randomly capitalize letters in a body of text. please do not confuse this disease with a blatant attempt at steganography.

Reveals: MEET AT THE FRONT OF THE TRAP

Hiding a Message inside Text

- First-letter algorithm
- Every n-th character
- Altering the amount of whitespace
- Using a publicly available cover source

Hiding a Message inside Images

- The most popular medium!
- ***Least-significant bit (LSB) modifications***
 - 24-bit vs. 8-bit images
 - Tools to implement LSB: EzStego and S-Tools
- ***Masking and Filtering***
- ***Algorithms and Transformations***

Hiding an Image within an Image

- Removing all but the two least significant bits of each color component produces an almost completely black image. Making that image 85 times brighter produces the image below



- source: wikipedia.org

Hiding a Message inside Audio or Videos Files

- **Advantages**

- Human limitations – 20.000 Hz.
- Large amount of data that can be hidden inside
- Hard to recognize because of because of the continuous flow of information (moving stream of images and sound)

Steganalysis

- Detecting and decoding the hidden data within a given medium
- Even if secret content is not revealed, modifying the cover medium changes the medium's statistical properties
- Distributed Dictionary Attack

POSSIBLE USES OF STEGANOGRAPHY	DRAWBACKS
■ Used to combine explanatory information with an image (like doctor's notes accompanying an X-ray)	■ Could accidentally degrade or render an image misleading
■ Embedding corrective audio or image data in case corrosion occurs from a poor connection or transmission	■ Could counteract and be counterproductive with the original image
■ Peer-to-peer private communications	■ Doesn't hide the fact that an e-mail was sent, negating the purpose of secret communications
■ Posting secret communications on the Web to avoid transmission	■ Someone else with a steganography detection and cracking tool could expose the message
■ Copyright protection	■ A form of this already exists, called digital watermarking, but requires use of separate hardware tools because steganographic software can't use separate hardware tools. Steganographic software also can't protect the watermark.
■ Maintaining anonymity	■ Easier to open free Web-based e-mail or use cloaked e-mail
■ Hiding data on the network in case of a breach	■ Better to understand and effectively use standardized encryption

Future of Steganography

- The most important use of steganographic techniques will probably lie in the field of digital watermarking
- Might become limited under laws - governments claimed that criminals use steganography to communicate

...the End